Linear Regression

$$Y = mX + b + \varepsilon$$

$$E[Y] = mE[X] + b$$
$$Var(Y) = m^2 Var(X) + Var(\varepsilon)$$

$$\text{Signal to Noise Ratio (SNR)} = \frac{m^2 Var(X)}{Var(\varepsilon)}$$

$$\rho = \frac{Cov(X,Y)}{\sqrt{Var(X)\,Var(Y)}} = \pm\sqrt{\frac{1}{1+SNR^{-1}}}$$

$$Cov(X,Y) = Cov(X, mX+\varepsilon) = m\,Cov(X,X) + \underbrace{Cov(X,\varepsilon)}_{0} = m\,Var(X)$$

So, $\boxed{\hat{m} = \frac{Cov(X,Y)}{Var(X)}}$, $\hat{b} = E[Y] - \frac{Cov(X,Y)}{Var(X)} E[X]$ $\longleftrightarrow$ Solving $\hat{m}, \hat{b} = \underset{m,b}{\arg\min}\,(y - mX)^2$

$$m = \frac{Cov(X,Y)}{Var(X)} = a \cdot \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}} \Rightarrow a = \frac{\sqrt{Var(Y)}}{\sqrt{Var(X)}} \Rightarrow \hat{m} = \rho \cdot \frac{SD(Y)}{SD(X)}$$

Then, if $X^* = \frac{X}{SD(X)}$, $Y^* = \frac{Y}{SD(Y)}$

$$m^* = \frac{Cov(X^*, Y^*)}{Var(X^*)} = \frac{\frac{1}{\sqrt{Var(X)Var(Y)}}\,Cov(X,Y)}{\cancel{\frac{1}{Var(X)}\,Var(X)}} = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}} = \rho$$

So, scaling by X, Y by $\frac{1}{SD(X)}$, $\frac{1}{SD(Y)}$ can make $m^* = \rho$.

Bias - Variance - Noise Decomposition of 1-D OLS

$$E[(y_i - \hat{y}_i)^2] = Bias(\hat{m})^2 + Bias(\hat{b})^2 + Var(\hat{m}) + Var(\hat{b}) + Var(\varepsilon)$$

Multiple Linear Regression

In 1-D OLS,

- $y = mX + b + \varepsilon$
- $\hat{m} = \dfrac{Cov(X,Y)}{Var(X)} = \rho \cdot \dfrac{SD(Y)}{SD(X)}$

If $X^* = \dfrac{1}{\sqrt{SD(X)}}$, $Y^* = \dfrac{1}{\sqrt{SD(Y)}}$, then $\hat{m} = \rho$.

For multiple linear regression:

$$\underset{n\times 1}{Y} = \underset{n\times p}{X}\,\underset{p\times 1}{\beta} + \underset{n\times 1}{\varepsilon}$$

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} | & & | \\ x_1 & \cdots & x_p \\ | & & | \end{bmatrix} \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

- Now, $\beta \neq Corr(x_{i \in \{1,\dots,p\}}, y)$
  - Covariants between predictors $Cov(x_i, x_j) \neq 0$
  - Lower $\beta_i \neq$ higher correlation

$$\hat{\beta} = \underset{\beta}{\arg\min} \, |y - X\beta|_2^2 \Rightarrow \hat{\beta} = (X^T X)^{-1} X^T y$$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y}_i)^2}$$

- Measures how well our model does compared to the baseline model $\bar{y}$.

Ridge Regression

- Adds a L2-regularizer to smooth training
- Can give a solution even if X doesn't have linearly independent rows
- Smoothing spreads out decision boundaries so one data point is not as influential $\Rightarrow$ balance between training/not training the data

$$\min_{\beta} \; |y - X\beta|_2^2 + \alpha |\beta|_2^2$$

$$\Rightarrow \quad \hat{\beta} = (X^T X + \alpha I)^{-1} X^T y$$

Auto Regression

- Use labels from $t-1$ to predict labels at t

AR(1) model:

$$\hat{y}_t = X\hat{\beta} + y_{t-1}\hat{\beta}_{t-1}$$

- Longitudinal Model (L) – use $y_{t-1}, y_{t-2}, \ldots$ to predict y_t
- Cross-sectional model (C) – use predictors $x_1 \ldots x_p$ to predict y_t.

Inductive vs. Transductive Models

Inductive:
- Given (x, y) output model $f(\theta)$
- Tries to learn a global rule from training data, and apply it to test data
- Generalizable

Transductive: Learn specific patterns about this particular set of unlabeled data

- Look for specific patterns in this dataset only, use training data as guide
- E.g. Data structure that organizes the x, y pairs

  To find y and given x_{new}, query into the data structure and use the closest (x, y) pair.